CRBench: A Method-Agnostic, Resource-Aware Evaluation Framework for Long-Context Large Language Models

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Abstract

As Large Language Models (LLMs) expand context windows beyond 32K, 128K, and 1M tokens, the Key-Value (KV) cache becomes the primary operational bottleneck during autoregressive inference, consuming tens of gigabytes of GPU memory. While diverse compression paradigms have emerged—spanning KV quantization, attention-sink eviction, semantic token merging, and low-rank projection—evaluation remains fragmented: existing benchmarks measure accuracy without resource tracking, while compression literature reports ad-hoc memory ratios on unstandardized tasks. In this work, we present **CRBench** (*Context Resource Benchmark*), a method-agnostic, query-level evaluation framework that rigorously quantifies task quality relative to memory and system overheads. CRBench formalizes the atomic evaluation primitive as the 4-tuple $(\mathcal{M}, q, \mathcal{R}_{\text{dense}}, \mathcal{A})$, logging every candidate run alongside an uncompressed FP16 reference. We formulate canonical absolute task quality $Q_{\text{abs}} \in [0, 100]$, exact analytical and physical resource savings $R_{\text{mem}} \in [0, 100]$ via effective bits-per-element b_{eff}, and a composite utility score $\mathcal{S}_{\text{res}} = \alpha Q_{\text{abs}} + (1-\alpha) R_{\text{mem}}$ (default $\alpha = 0.70$), complemented by a decoupled Part 2 system serving track ($\mathcal{S}_{\text{sys}}$) capturing prefill latency (TTFT), decode throughput, and peak VRAM.

We evaluate 8 context compression methods across 3 foundation models (Gemma 4 E2B, Gemma 4 E4B, and Qwen2.5 7B) spanning 3,150 queries across 5 long-context tasks from 2K to 128K tokens on NVIDIA RTX GPU hardware. Empirical results show that: (1) KIVI-style INT4 quantization attains the highest cross-model system score ($\bar{\mathcal{S}}_{\text{sys}} = 70.3$) via near-zero latency overhead ($\phi \geq 0.96\times$); (2) SnapKV delivers strong algorithmic utility ($\mathcal{S}_{\text{res}} = 60.8$) but suffers severe prefill hardware penalties ($\phi = 0.20\times$, TTFT= 83.4s on E4B) during un-fused attention pooling; (3) DKV-Mid achieves the lowest footprint (5.00–5.84 bpt) among high-accuracy methods; and (4) all methods fail catastrophically on multi-step variable tracking, identifying a fundamental base-model reasoning bottleneck. CRBench is open-sourced with an extensible adapter API.

1 Introduction

Modern Large Language Models (LLMs) have achieved remarkable breakthroughs across multi-document synthesis, complex multi-hop question answering, repository-scale code generation, and autonomous agent workflows (Brown et al., 2020; Achiam et al., 2023; Touvron et al., 2023; Bai et al., 2023a). These capabilities rely on long-context processing, expanding context horizons from 2,048 tokens to 32,768, 131,072, and beyond (Reid et al., 2024; DeepSeek-AI, 2024).

However, contextual expansion is fundamentally bounded by the memory footprint of the Key-Value (KV) cache in standard Transformer self-attention architectures (Vaswani et al., 2017). Because standard attention caches intermediate key and value tensor representations for every historical token across all layers and attention heads, KV cache memory scales linearly with sequence length L and batch size B:

$$M_{\text{KV}} = 2 \times N_{\text{layers}} \times N_{\text{kv_heads}} \times d_{\text{head}} \times L \times B \times \text{sizeof(dtype)} \quad \text{[bytes]} \tag{1}$$

For a representative 8-billion parameter model (e.g., Llama-3.1-8B with 32 layers, 8 Key-Value heads under Grouped-Query Attention, and head dimension 128 in 16-bit precision), a single batch-1 128K-token sequence ($L = 131{,}072$) requires exactly 16.0 GiB (17.18 GB) of memory purely for the KV cache ($2 \times 32 \times 8 \times 128 \times 131{,}072 \times 2$ bytes)—exceeding the model's static weight footprint and exhausting consumer GPU VRAM.

To address this severe memory wall, the machine learning community has introduced a diverse taxonomy of context compression paradigms:

1. **KV Cache Quantization**: Low-precision numerical formats (FP8, INT8, INT4, INT2) with per-channel or group-wise dynamic scaling (Dettmers et al., 2022; Hooper et al., 2024; Liu et al., 2024).
2. **KV Cache Eviction & Sparsification**: Attention sinks and dynamic heavy-hitter token eviction (e.g., StreamingLLM (Xiao et al., 2023), SnapKV (Li et al., 2024), H_2O (Zhang et al., 2023), Scissorhands (Liu et al., 2023)).
3. **KV Cache Merging & Pooling**: Temporal and semantic clustering of adjacent token representations (Ge et al., 2023; Zhang et al., 2024a).
4. **Low-Rank & Subspace Projection**: SVD and principal-component latent projections (Sharma et al., 2024).
5. **Factorized State Memory**: Shared static subspace bases with sparse dynamic updates.

Despite this rapid algorithmic progress, an acute methodological challenge exists: *the field lacks a unified, method-agnostic, resource-aware evaluation benchmark.* Current long-context benchmarks (e.g., Needle-In-A-Haystack (Kamradt, 2023), RULER (Hsieh et al., 2024), LongBench (Bai et al., 2023b), LongBench v2 (Bai et al., 2024)) evaluate task accuracy under uncompressed representations without tracking memory consumption. Conversely, compression papers report accuracy on idiosyncratic task subsets alongside theoretical compression ratios, without controlling for base model capability, metadata storage overheads, or hardware latency.

CRBench Overview. CRBench resolves these challenges with a query-level, dense-paired evaluation primitive that jointly records task outcomes, memory, and runtime. As illustrated in Figure 1, CRBench decouples quality–memory utility (Part 1, $\mathcal{S}_{\text{res}}$) from hardware system serving utility (Part 2, $\mathcal{S}_{\text{sys}}$).

Method Neutrality Disclosure. One of the evaluated methods (Differential KV / DKV) was introduced by the authors of this work. To maintain strict evaluation neutrality and prevent benchmarking bias, CRBench is designed as a method-agnostic, open-source framework with standardized, identical execution pipelines across all candidate adapters, fully public harness code, and transparent reproducibility across all evaluated configurations.

Core Contributions. The primary contributions of this work are:

- **The Query-Level Evaluation Primitive**: We formulate benchmark evaluation around the atomic 4-tuple $(\mathcal{M}, q, \mathcal{R}_{\text{dense}}, \mathcal{A})$, pairing every compressed method evaluation directly against the identical model's uncompressed dense FP16 reference on the exact same prompt.
- **Auditable Quality and Dense Pairing**: We report canonical absolute task quality $Q_{\text{abs}} \in [0, 100]$ while preserving the paired dense score and paired-execution coverage for every query.
- **Explicit Analytical & Physical Memory Accounting (R_{mem})**: We formalize effective bits-per-element (b_{eff}), accounting not only for raw tensor payloads but also quantization scales, codebooks, eviction index bitmaps, and alignment overheads.
- **Desiderata-Based Scoring Formulation ($\mathcal{S}_{\text{res}}$)**: We audit seven candidate utility formulations against five formal measurement axioms, establishing the linear additive formulation $\mathcal{S}_{\text{res}} = \alpha Q + (1 - \alpha) R_{\text{mem}}$ as a transparent, non-destructive scoring metric.

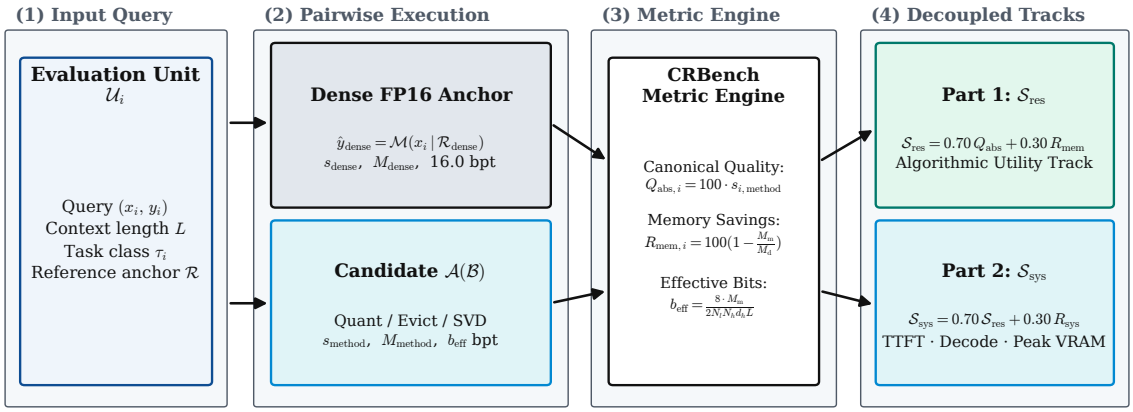


Figure 1: **CRBench Method-Agnostic Evaluation Framework Architecture.** The benchmark decouples evaluation into two complementary tracks: Part 1 (Algorithmic Representation Utility $\mathcal{S}_{\text{res}}$) and Part 2 (Hardware System Track $\mathcal{S}_{\text{sys}}$). Every candidate context representation $\mathcal{A}(\mathcal{B})$ is evaluated pairwise against the base model's uncompressed dense reference $\mathcal{R}_{\text{dense}}$ on the identical input query (x_i, y_i).

Table 1: Comparison of Evaluation Paradigms vs. CRBench. Summary of dimensions captured across long-context benchmarks, synthetic needle tests, inference profilers, and CRBench.

Evaluation Paradigm	Quality	Memory	Latency	Dense Anchor	Pairwise
Long-Context Suites (LongBench, L-Eval)	✓	✗	✗	✗	✗
Synthetic Needle Tests (NIAH, RULER)	✓	✗	✗	✗	✗
Inference Systems Profilers (vLLM)	✗	✓	✓	✗	✗
Ad-hoc Compression Literature	✓	△	△	✗	✗
CRBench (This Work)	✓	✓	✓	✓	✓

- **Decoupled Two-Track Architecture**: We decouple algorithmic representation quality (Part 1) from hardware-dependent runtime profiling (Part 2, $\mathcal{S}_{\text{sys}}$), capturing TTFT, decode throughput, peak VRAM, and system multiplier ϕ.
- **Cross-Model Publication Benchmark (3,150 Queries)**: We evaluate 8 methods across Gemma 4 E2B (2K–128K ladder), Gemma 4 E4B (2K–32K), and Qwen2.5 7B (2K–32K) on NVIDIA RTX hardware.

2 Related Work

2.1 Limitations of Existing Evaluation Paradigms

Existing approaches to evaluating long-context models and KV compression techniques suffer from structural shortcomings, summarized in Table 1:

1. **Quality–Resource Disconnection**: Traditional benchmarks (e.g., L-Eval (An et al., 2023), BAMBOO (Dong et al., 2023), InfiniteBench (Zhang et al., 2024b)) measure accuracy without tracking memory. A method achieving 90% accuracy while consuming 16 bits per token is treated identically to a method achieving 90% accuracy at 4 bits per token, despite the latter offering 4× greater hardware efficiency.
2. **Cross-Model Confounding**: Absolute task accuracy also reflects the base model's native capability. CRBench therefore reports each model separately and logs paired dense outcomes on the identical query, allowing readers to inspect representation effects without presenting a cross-model score as a causal comparison.

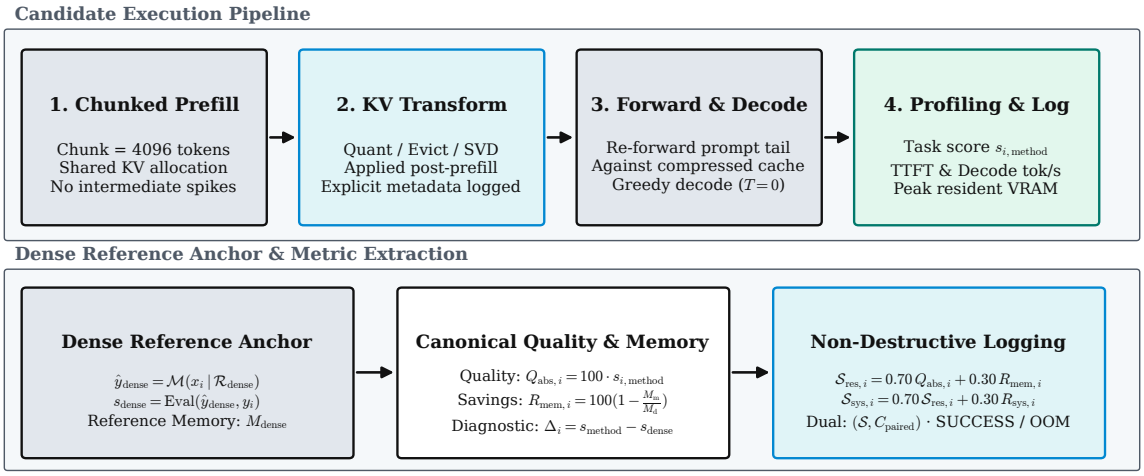


Figure 2: **Atomic Query-Level Evaluation Pipeline in CRBench.** Execution flow from a single query through chunked prefill, transformed-cache re-forwarding, exact metadata tracking, paired dense diagnostics, canonical quality–memory scoring, and dataset-level aggregation.

3. **Omission of Metadata and Runtime Overheads**: Many compression works report theoretical payload bits while omitting the memory required for scaling factors, outlier vectors, or eviction index tables. Furthermore, algorithms that save memory via uncoalesced memory gathers or CPU–GPU synchronization can severely degrade prefill latency (TTFT) and decode throughput.

3 Problem Definition & Mathematical Formulation

3.1 The Atomic Evaluation Primitive

Let $\mathcal{M}$ denote an autoregressive language model parameterized by weights Θ. An evaluation dataset $\mathcal{D} = \{(x_i, y_i, \tau_i, L_i)\}_{i=1}^{N}$ consists of queries x_i, ground-truth target outputs y_i, task types τ_i, and prompt context token lengths L_i.

Let $\mathcal{A}$ denote a candidate context memory adapter (e.g., INT4 quantization or SnapKV eviction) operating under assigned resource budget $\mathcal{B}$. In standard dense inference, the uncompressed reference adapter is denoted $\mathcal{R}_{\text{dense}}$ (16-bit FP16/BF16 representation).

Definition 1 (CRBench Atomic Evaluation Unit)**.** The fundamental evaluation unit in CRBench is the 4-tuple:

$$\mathcal{U}_i = \Big(\mathcal{M},\ (x_i, y_i),\ \mathcal{R}_{\text{dense}},\ \mathcal{A}(\mathcal{B})\Big) \tag{2}$$

For every query (x_i, y_i), CRBench executes:

1. **Dense Reference Execution**: Model $\mathcal{M}$ generates prediction $\hat{y}_{i,\text{dense}} = \mathcal{M}(x_i \mid \mathcal{R}_{\text{dense}})$.
2. **Candidate Adapter Execution**: Model $\mathcal{M}$ generates prediction $\hat{y}_{i,\text{method}} = \mathcal{M}(x_i \mid \mathcal{A}(\mathcal{B}))$.
3. **Quality Measurement**: Raw task scores $s_{i,\text{dense}} = \text{Eval}(\hat{y}_{i,\text{dense}}, y_i) \in [0,1]$ and $s_{i,\text{method}} = \text{Eval}(\hat{y}_{i,\text{method}}, y_i) \in [0,1]$ are recorded.
4. **Resource Measurement**: Memory footprints $M_{i,\text{dense}}$ and $M_{i,\text{method}}$ (in bytes) are computed analytically and tracked physically.

3.2 Canonical Quality and Dense-Reference Diagnostics

The canonical leaderboard deliberately uses an absolute task-success quantity. This makes scores comparable across the full suite and keeps a weak dense baseline from inflating a compressed method's apparent retention.

The paired dense execution remains essential: it records the reference outcome, enables failure diagnosis, and defines coverage.

Definition 2 (Canonical Absolute Task Quality)**.** Let $s_{i,\text{method}} \in [0,1]$ be the task evaluator's score for a completed candidate run. Its canonical quality is:

$$Q_{\text{abs},i} = 100.0 \, \min\left(1.0, \max\left(0.0, s_{i,\text{method}}\right)\right). \tag{3}$$

Paired diagnostics and coverage. We log the signed paired difference $\Delta_i = s_{i,\text{method}} - s_{i,\text{dense}}$ and report paired execution coverage $C_{\text{paired}} = N_{\text{paired-success}}/N_{\text{total}}$. Thus, a zero score from a completed run is distinguished from an out-of-memory or runtime failure. Relative retention can be recomputed from the logged pairs for a task with a defensible chance floor, but it is not substituted into the canonical reported scores.

3.3 Memory Resource Accounting (R_{mem})

CRBench enforces explicit state memory accounting. For any adapter $\mathcal{A}$, the total memory consumed by the context representation is partitioned into three additive components:

$$M_{\text{method}}(L) = M_{\text{algo}}(L) + M_{\text{meta}}(L) + M_{\text{align}}(L) \quad \text{[bytes]} \tag{4}$$

where M_{algo} is the raw compressed tensor payload, M_{meta} represents all mandatory auxiliary metadata (quantization scale factors, zero-point offsets, codebook dictionaries, or 32-bit integer token position index arrays for eviction methods), and M_{align} represents hardware memory alignment and padding overheads.

Definition 3 (Effective Bits Per KV Element)**.** The effective bits-per-element (b_{eff}) measures the total memory footprint normalized by the total uncompressed Key-Value tensor scalar count ($2 \times N_{\text{layers}} \times N_{\text{kv_heads}} \times d_{\text{head}} \times L$):

$$b_{\text{eff}} = \frac{8 \cdot M_{\text{method}}(L)}{2 \cdot N_{\text{layers}} \cdot N_{\text{kv_heads}} \cdot d_{\text{head}} \cdot L} \quad \text{[bits/element]} \tag{5}$$

For standard uncompressed 16-bit float KV caches (FP16/BF16), $b_{\text{eff}} = 16.0$ bits/element.

Definition 4 (Memory Resource Savings)**.** The normalized memory resource savings $R_{\text{mem},i} \in [0.0, 100.0]$ is:

$$R_{\text{mem},i} = 100.0 \times \max\left(0.0,\ 1.0 - \frac{M_{i,\text{method}}}{M_{i,\text{dense}}}\right) \tag{6}$$

4 CRBench Scoring Formulation

4.1 Measurement Desiderata for Utility Formulation

A useful scalar utility must be monotone in both absolute task quality and resource savings, bounded on $[0,100]^2$, and Pareto consistent. The linear utility below satisfies these properties directly; its short proof and a comparison with alternative formulations are deferred to the supplementary material.

4.2 Part 1: Resource Utility Score ($\mathcal{S}_{\text{res}}$)

CRBench defines the canonical Part 1 Resource Score using a linear additive utility formulation:

$$\mathcal{S}_{\text{res},i} = \alpha \cdot Q_{\text{abs},i} + (1-\alpha) \cdot R_{\text{mem},i} \tag{7}$$

where $\alpha \in [0.10, 0.90]$ is the global quality preference weight, with standard configuration $\alpha = 0.70$.

Default trade-off. The $\alpha = 0.70$ configuration makes a one-point gain in task quality worth 2.33 points of memory savings. A dense run with perfect task success has score 70.0, whereas a 4-bit method at 94% task quality and 75% savings has score 88.3. The sensitivity analysis is supplied in the supplement.

4.3 Dataset Aggregation & Coverage Reliability (C_{paired})

For an evaluation dataset $\mathcal{D}$, the aggregate benchmark score is computed by averaging query-level utility scores:

$$\bar{\mathcal{S}}_{\text{res}} = \frac{1}{|\mathcal{D}|}\sum_{i=1}^{|\mathcal{D}|}\mathcal{S}_{\text{res},i}, \quad \bar{Q}_{\text{abs}} = \frac{1}{|\mathcal{D}|}\sum_{i=1}^{|\mathcal{D}|}Q_{\text{abs},i}, \quad \bar{R}_{\text{mem}} = \frac{1}{|\mathcal{D}|}\sum_{i=1}^{|\mathcal{D}|}R_{\text{mem},i} \tag{8}$$

CRBench computes 95% non-parametric bootstrap confidence intervals with $B = 2,000$ resamples.

Coverage Decoupling. CRBench reports paired query execution coverage $C_{\text{paired}} = \frac{N_{\text{paired_success}}}{N_{\text{total}}} \in [0.0, 1.0]$ alongside $\mathcal{S}_{\text{res}}$ as a dual-metric pair $(\mathcal{S}_{\text{res}}, C_{\text{paired}})$, rather than multiplicatively folding coverage into $\mathcal{S}_{\text{res}}$. This preserves algorithmic evaluation integrity even if occasional memory allocator faults occur.

5 System Evaluation Track (Part 2)

While Part 1 evaluates algorithmic memory representation efficiency, practical deployment requires analyzing physical latency and throughput. An adapter that compresses tensors effectively but relies on un-fused loops or irregular memory gathers can suffer severe latency degradation.

5.1 Profiled System Metrics

CRBench incorporates hardware-synchronized wall-clock timers and device memory allocators to capture:

1. **Time-To-First-Token (TTFT)**: Total wall-clock time (in seconds) required to process prompt context of length L and emit the first token.
2. **Autoregressive Decode Throughput ($\mathcal{T}_{\text{dec}}$)**: Generation speed in tokens per second (tok/s):

$$\mathcal{T}_{\text{dec}} = \frac{N_{\text{generated}}}{\Delta t_{\text{decode}}} \quad [\text{tok/s}] \tag{9}$$

3. **Peak Resident Device VRAM ($\mathcal{M}_{\text{peak}}$)**: Maximum allocated and reserved GPU memory during inference.

5.2 System Efficiency and Part 2 Score ($\mathcal{S}_{\text{sys}}$)

CRBench defines composite system efficiency $R_{\text{sys}} \in [0.0, 100.0]$:

$$R_{\text{sys},i} = w_{\text{mem}} \cdot R_{\text{mem},i} + w_{\text{ttft}} \cdot R_{\text{ttft},i} + w_{\text{thru}} \cdot R_{\text{thru},i} \tag{10}$$

where $w_{\text{mem}} = 0.50, w_{\text{ttft}} = 0.25, w_{\text{thru}} = 0.25$, and the normalized runtime efficiency components are defined relative to the dense baseline:

$$R_{\text{ttft},i} = \min\left(100.0,\ 50.0 \times \frac{\text{TTFT}_{i,\text{dense}}}{\text{TTFT}_{i,\text{method}}}\right), \quad R_{\text{thru},i} = \min\left(100.0,\ 50.0 \times \frac{\mathcal{T}_{i,\text{method}}}{\mathcal{T}_{i,\text{dense}}}\right) \tag{11}$$

The composite **CRBench Part 2 System Score** then extends the Part 1 score with system efficiency:

$$\mathcal{S}_{\text{sys},i} = \alpha \cdot \mathcal{S}_{\text{res},i} + (1 - \alpha) \cdot R_{\text{sys},i}. \tag{12}$$

We define the overall system multiplier $\phi \in [0.0, 1.0]$:

$$\phi = \min\left(1.0,\ \frac{\text{TTFT}_{\text{dense}}}{\text{TTFT}_{\text{method}}} \cdot \frac{\text{VRAM}_{\text{dense}}}{\text{VRAM}_{\text{method}}}\right) \tag{13}$$

Methods with $\phi < 1.0$ incur hardware penalties from prolonged prefill or inflated peak memory.

6 Experimental Setup

6.1 Evaluated Models & Context Ladders

We evaluate three open-weight foundation models across 3,150 standardized queries:

- **Gemma 4 E2B**: Evaluated across the complete 2K $\rightarrow$ 128K context ladder (2K, 8K, 32K, 64K, 128K tokens, 1,050 queries).
- **Gemma 4 E4B**: Evaluated across context horizons from 2K to 32K (2K, 4K, 8K, 16K, 32K tokens, 1,050 queries).
- **Qwen2.5 7B**: Evaluated across context horizons from 2K to 32K (2K, 4K, 8K, 16K, 32K tokens, 1,050 queries).

6.2 Candidate KV Compression Methods

Published algorithms are driven directly from vendored upstream repositories:

1. **KIVI / KV-Quant**: 4-bit per-channel keys, 4-bit per-token values (Liu et al., 2024).
2. **SnapKV**: Observation-guided heavy-hitter eviction ($L_{\text{obs}} = 32, k = 7$) (Li et al., 2024).
3. **DKV-Mid / High**: Differential KV with 128- / 256-rank subspace and dynamic residuals.
4. **Dense FP16**: Full uncompressed 16-bit reference baseline ($b_{\text{eff}} = 16.0$ bpt).
5. **Low-Rank KV**: Per-head mean-centered truncated SVD projection (Sharma et al., 2024).
6. **StreamingLLM**: Attention sink ($N_{\text{sink}} = 4$) + rolling local window eviction (Xiao et al., 2023).
7. **KV-Merging**: Temporal adjacent token mean-pooling (Ge et al., 2023).

6.3 Evaluation Tasks

1. **Single & Multi NIAH**: Single- and distributed multi-target needle retrieval at parameterized insertion depths.
2. **RULER KV**: High-entropy dictionary key-value lookup amid distracting pairs (Hsieh et al., 2024).
3. **Variable Tracking**: Multi-step variable reassignment tracking (Hsieh et al., 2024).
4. **Multi-Hop QA**: Cross-document synthetic multi-hop reasoning.

7 Results

7.1 Cross-Model Algorithmic Representation Fidelity ($\mathcal{S}_{\text{res}}$)

Table 2 presents the Part 1 leaderboard across all three models under canonical scoring $\mathcal{S}_{\text{res}} = 0.70 \cdot Q_{\text{abs}} + 0.30 \cdot R_{\text{mem}}^{\text{agg}}$.

7.2 Cross-Model Hardware System Serving Utility ($\mathcal{S}_{\text{sys}}$)

Table 3 presents the Part 2 system scores. The per-model distribution is provided in the supplementary material.

7.3 Task-Wise Performance Breakdown

Figure 4 visualizes the cross-model mean accuracy by task archetype.

Table 2: **Cross-Model Part 1 ($\mathcal{S}_{\text{res}}$) Leaderboard.** Evaluated under canonical Option B physical memory aggregation across 3,150 queries. Best score per model in **bold**.

	Gemma 4 E2B (2K–128K)			Gemma 4 E4B (2K–32K)			Qwen2.5 7B (2K–32K)		
Method	**Accuracy**	$b_{\text{eff}}^{\text{agg}}$	$\mathcal{S}_{\text{res}}$	**Accuracy**	$b_{\text{eff}}^{\text{agg}}$	$\mathcal{S}_{\text{res}}$	**Accuracy**	$b_{\text{eff}}^{\text{agg}}$	$\mathcal{S}_{\text{res}}$
KIVI / KV-Quant	46.4%	6.01 bpt	51.2	62.0%	6.04 bpt	**62.1**	60.7%	6.04 bpt	**61.1**
SnapKV	46.8%	6.00 bpt	**51.5**	60.0%	6.00 bpt	60.8	58.2%	6.00 bpt	59.5
DKV-Mid	44.7%	5.46 bpt	51.0	55.3%	5.00 bpt	59.4	53.3%	5.84 bpt	56.3
DKV-High	46.7%	10.78 bpt	42.5	61.3%	10.68 bpt	52.9	59.1%	12.14 bpt	48.6
Dense FP16	46.4%	16.00 bpt	32.5	64.0%	16.00 bpt	44.8	60.0%	16.00 bpt	42.0
Low-Rank KV	31.2%	6.07 bpt	40.5	48.8%	6.27 bpt	52.4	28.6%	6.09 bpt	38.6
StreamingLLM	22.9%	6.00 bpt	34.8	25.5%	6.00 bpt	36.6	36.5%	6.00 bpt	44.3
KV-Merging	13.3%	6.01 bpt	28.0	25.3%	6.03 bpt	36.4	19.9%	6.03 bpt	32.6

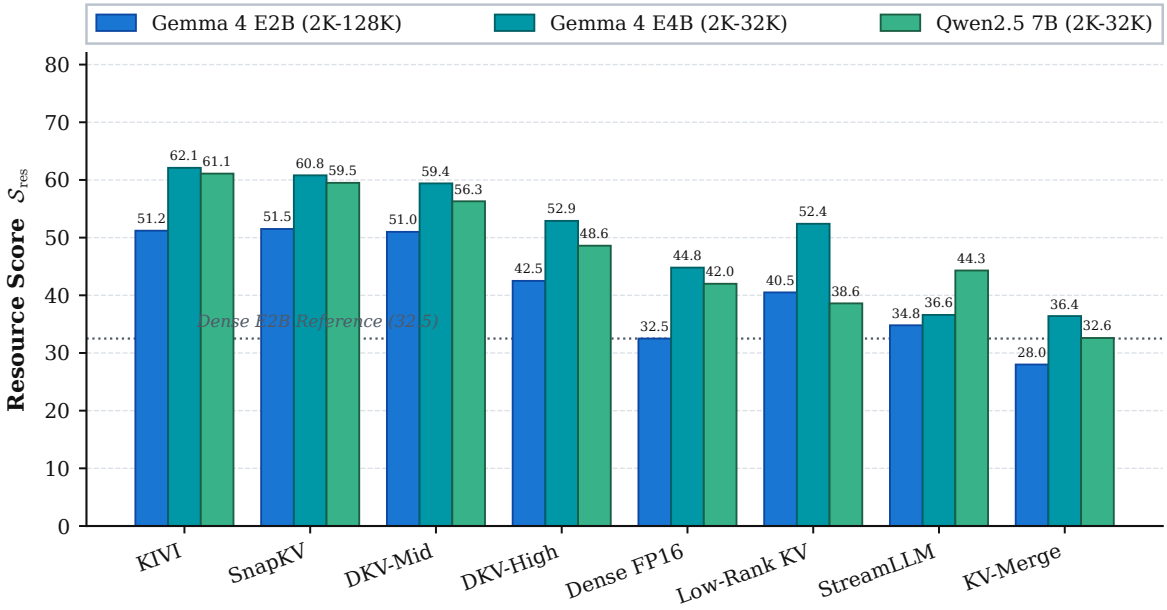


Figure 3: **Cross-Model Part 1 Resource Utility ($\mathcal{S}_{\text{res}}$) Comparison.** Evaluated across Gemma 4 E2B, Gemma 4 E4B, and Qwen2.5 7B.

7.4 Ultra-Long Context Ladder (2K → 128K) Scaling

Figure 5 traces performance across the full 2K–128K context ladder on Gemma 4 E2B, illustrating the degradation cliff beyond 64K tokens.

7.5 Empirical Quality–Memory Pareto Frontier

Figure 6 maps task accuracy against effective bits-per-element b_{eff}, highlighting the non-dominated Pareto frontier across all three models.

8 Analysis & Discussion

KIVI Dominates System Serving ($\bar{\mathcal{S}}_{\text{sys}} = 70.3$). KIVI-style INT4 quantization achieves the highest cross-model system score because asymmetric quantization operates during cache writes without prefill latency spikes, maintaining $\phi \geq 0.96\times$ across all models.

Table 3: **Cross-Model Part 2 Hardware System Serving Utility ($\mathcal{S}_{\text{sys}}$).** Ranked by cross-model mean score.

Method	Gemma 4 E2B	Gemma 4 E4B	Qwen2.5 7B	Cross-Model Mean $\mathcal{S}_{\text{sys}}$
KIVI / KV-Quant	64.6	73.4	72.8	**70.3**
DKV-Mid	59.1	66.3	54.2	**59.9**
Low-Rank KV	57.4	65.9	54.9	**59.4**
Dense FP16	52.8	61.4	59.4	**57.8**
StreamingLLM	54.3	55.6	61.0	**57.0**
SnapKV	60.6	48.6	53.7	**54.3**
DKV-High	52.7	60.0	48.5	**53.7**
KV-Merging	49.6	55.5	52.8	**52.6**

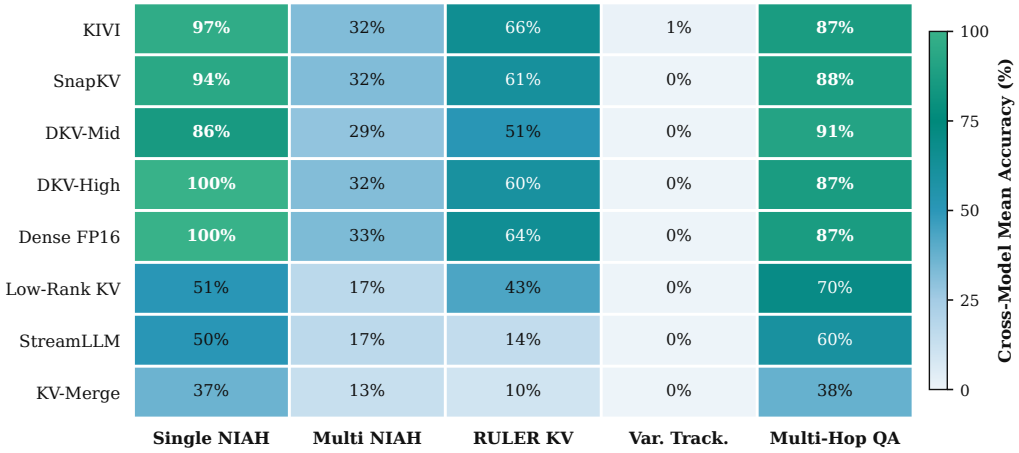


Figure 4: **Task-Wise Accuracy Heatmap Across Compression Paradigms.** Cross-model average accuracy (%) across all three evaluated models.

SnapKV's Prefill Hardware Penalty. While SnapKV attains high algorithmic utility ($\mathcal{S}_{\text{res}} = 60.8$ on E4B), un-fused observation pooling on 32K sequences inflates TTFT to 83.4s and peak VRAM to 2,632 MB ($\phi = 0.20\times$), dropping its Part 2 score to 48.6 and confirming the necessity of decoupled system profiling.

DKV-Mid Pushes the Memory Frontier (5.00–5.84 bpt). DKV-Mid achieves the lowest footprint among accurate methods (5.00 bpt on E4B) and leads Multi-Hop QA (91.1%). However, SVD projection increases TTFT (22.3s on E2B, 39.6s on 7B), resulting in $\phi \approx 0.49$–$0.83\times$.

Variable Tracking: Inherent Model Failure. All 8 methods—including uncompressed dense FP16—score near 0.0% on Variable Tracking, reflecting an intrinsic architectural limitation in multi-step pointer dereferencing rather than compression error.

9 Limitations & Research Scope

We identify four key operational boundaries: (1) **Model Scale**: Current evaluation benchmarks 2B to 7B architectures; future extensions will evaluate 13B+ and 70B+ model families. (2) **Task Coverage**: Our suite prioritizes long-context needle retrieval, associative lookup, and multi-hop reasoning; open-ended agentic dialogues and summarization represent natural next steps. (3) **Hardware Scope**: Profiling is conducted on single-node workstation GPUs; distributed multi-GPU serving may alter Part 2 communication costs. (4) **Kernel Fusion**: Algorithms evaluated via pure PyTorch incur prefill memory and dispatch latency penalties that dedicated fused CUDA/Triton kernels can substantially mitigate.

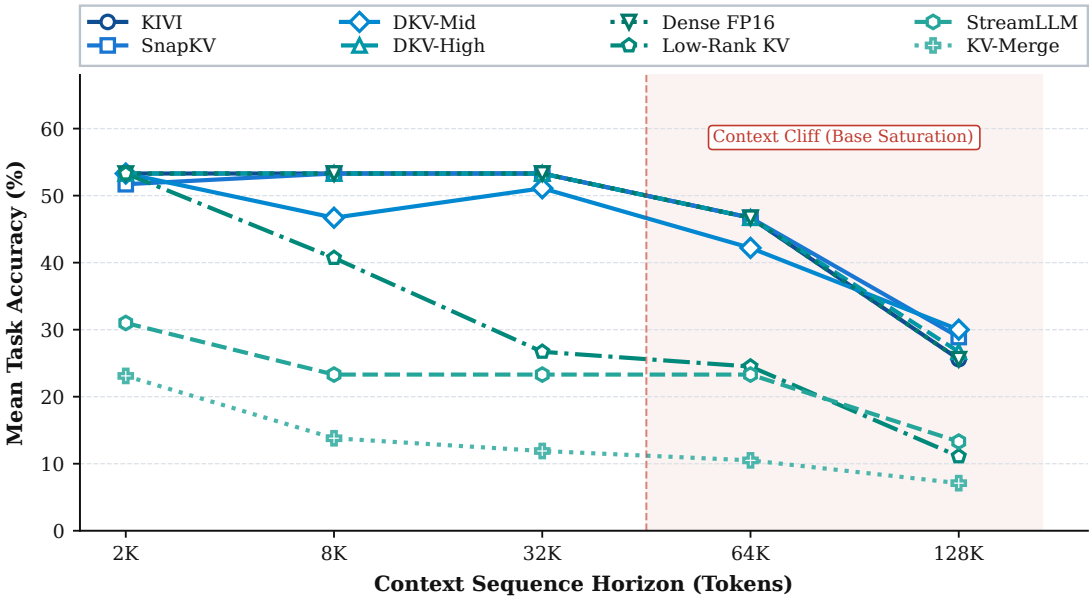


Figure 5: **Ultra-Long Context Ladder Scaling (2K to 128K Tokens) on Gemma 4 E2B.** Accuracy averaged across all 5 tasks, showing sharp degradation at 64K–128K shared with the uncompressed baseline.

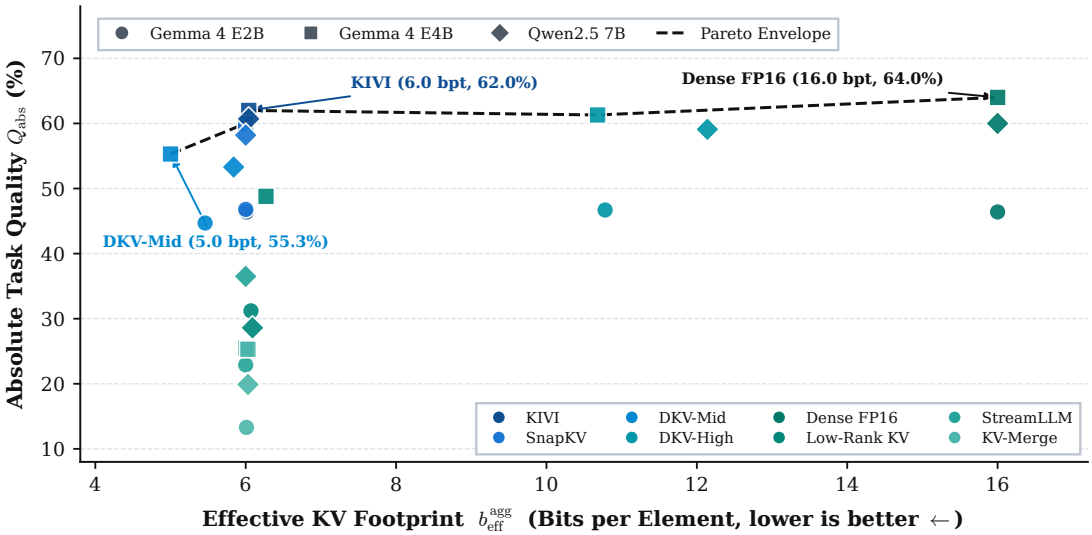


Figure 6: **Quality–Memory Pareto Frontier Across 3 Foundation Models.** Evaluated across 24 compression configurations, mapping task accuracy against effective bits-per-element b_{eff}.

10 Conclusion

We presented **CRBench**, a method-agnostic, resource-aware evaluation framework for long-context LLMs. By anchoring evaluations pairwise against the uncompressed dense reference, enforcing explicit memory accounting via b_{eff}, and decoupling algorithmic utility ($\mathcal{S}_{\text{res}}$) from hardware serving ($\mathcal{S}_{\text{sys}}$), CRBench establishes a rigorous foundation for evaluating context memory representations.

Broader Impact Statement

Context representation and compression technologies directly influence the economic accessibility and environmental footprint of frontier artificial intelligence. By reducing KV cache memory demands by up to 3.2×,

efficient context representations democratize long-context serving on commodity hardware and substantially decrease data center energy consumption and carbon emissions. However, aggressive eviction and lossy quantization can induce behavioral drift, hallucinations, or silent omission of needle facts in high-stakes domains (e.g., clinical retrieval or legal contract analysis). CRBench directly mitigates these risks by providing an auditable, multi-dimensional evaluation suite that rigorously exposes failure modes and hardware bottlenecks before compression algorithms are deployed in production systems.

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Supplementary Material: CRBench

A Method-Agnostic, Resource-Aware Evaluation Framework for Long-Context Large Language Models

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Paper under double-blind review

A Full Per-Model Leaderboards

This section provides complete Part 1 (Algorithmic Representation Fidelity $\mathcal{S}_{\text{res}}$) and Part 2 (Hardware System Serving Utility $\mathcal{S}_{\text{sys}}$) leaderboards for all three evaluated foundation model families across 3,150 queries.

A.1 Gemma 4 E2B Leaderboard (2K $\rightarrow$ 128K Context Ladder, 1,050 Queries)

Table 1: Gemma 4 E2B Part 1 ($\mathcal{S}_{\text{res}}$) Algorithmic Representation Leaderboard.

Rank	Method	Accuracy	$b_{\text{eff}}^{\text{agg}}$	VRAM Saved	$\mathcal{S}_{\text{res}}$
1	**SnapKV**	46.8%	6.00 bpt	62.5%	**51.5**
2	**KIVI**	46.4%	6.01 bpt	62.4%	51.2
3	**DKV-Mid**	44.7%	5.46 bpt	65.9%	51.0
4	**DKV-High**	46.7%	10.78 bpt	32.6%	42.5
5	**Low-Rank KV**	31.2%	6.07 bpt	62.1%	40.5
6	**StreamingLLM**	22.9%	6.00 bpt	62.5%	34.8
7	**Dense FP16** (Reference)	46.4%	16.00 bpt	0.0%	32.5
8	**KV-Merging**	13.3%	6.01 bpt	62.5%	28.0

Table 2: Gemma 4 E2B Part 2 ($\mathcal{S}_{\text{sys}}$) Hardware System Serving Leaderboard.

Rank	Method	$\mathcal{S}_{\text{sys}}$	$\mathcal{S}_{\text{res}}$	TTFT	Decode	Peak VRAM	ϕ
1	**KIVI**	**64.6**	51.2	12.51s	11.9 tok/s	847.0 MB	0.96×
2	**SnapKV**	60.6	51.5	13.04s	12.9 tok/s	2088.2 MB	0.82×
3	**DKV-Mid**	59.1	51.0	22.26s	13.8 tok/s	1389.9 MB	0.78×
4	**Low-Rank KV**	57.4	40.5	12.80s	12.3 tok/s	849.8 MB	0.97×
5	**StreamingLLM**	54.3	34.8	12.58s	13.7 tok/s	847.4 MB	1.00×
6	**Dense FP16**	52.8	32.5	13.18s	13.4 tok/s	1118.4 MB	1.00×
7	**DKV-High**	52.7	42.5	23.40s	13.8 tok/s	1391.0 MB	0.77×
8	**KV-Merging**	49.6	28.0	12.59s	13.7 tok/s	847.8 MB	1.00×

A.2 Gemma 4 E4B Leaderboard (2K $\rightarrow$ 32K Context Ladder, 1,050 Queries)

A.3 Qwen2.5 7B Leaderboard (2K $\rightarrow$ 32K Context Ladder, 1,050 Queries)

Table 3: Gemma 4 E4B Part 1 ($\mathcal{S}_{\text{res}}$) Algorithmic Representation Leaderboard.

Rank	Method	Accuracy	$b_{\text{eff}}^{\text{agg}}$	VRAM Saved	$\mathcal{S}_{\text{res}}$
1	**KIVI**	62.0%	6.04 bpt	62.2%	**62.1**
2	**SnapKV**	60.0%	6.00 bpt	62.5%	60.8
3	**DKV-Mid**	55.3%	5.00 bpt	68.7%	59.4
4	**DKV-High**	61.3%	10.68 bpt	33.3%	52.9
5	**Low-Rank KV**	48.8%	6.27 bpt	60.8%	52.4
6	**Dense FP16** (Reference)	64.0%	16.00 bpt	0.0%	44.8
7	**StreamingLLM**	25.5%	6.00 bpt	62.5%	36.6
8	**KV-Merging**	25.3%	6.03 bpt	62.3%	36.4

Table 4: Gemma 4 E4B Part 2 ($\mathcal{S}_{\text{sys}}$) Hardware System Serving Leaderboard.

Rank	Method	$\mathcal{S}_{\text{sys}}$	$\mathcal{S}_{\text{res}}$	TTFT	Decode	Peak VRAM	ϕ
1	**KIVI**	**73.4**	62.1	2.84s	12.8 tok/s	761.3 MB	1.00×
2	**DKV-Mid**	66.3	59.4	5.06s	12.3 tok/s	760.7 MB	0.83×
3	**Low-Rank KV**	65.9	52.4	3.18s	13.1 tok/s	761.5 MB	0.97×
4	**Dense FP16**	61.4	44.8	2.96s	12.4 tok/s	761.5 MB	1.00×
5	**DKV-High**	60.0	52.9	6.25s	12.2 tok/s	761.0 MB	0.76×
6	**StreamingLLM**	55.6	36.6	2.92s	13.6 tok/s	761.4 MB	1.00×
7	**KV-Merging**	55.5	36.4	2.91s	13.5 tok/s	761.5 MB	1.00×
8	**SnapKV**	48.6	60.8	83.38s	10.6 tok/s	2632.4 MB	0.20×

Table 5: Qwen2.5 7B Part 1 ($\mathcal{S}_{\text{res}}$) Algorithmic Representation Leaderboard.

Rank	Method	Accuracy	$b_{\text{eff}}^{\text{agg}}$	VRAM Saved	$\mathcal{S}_{\text{res}}$
1	**KV-Quant**	60.7%	6.04 bpt	62.2%	**61.1**
2	**SnapKV**	58.2%	6.00 bpt	62.5%	59.5
3	**DKV-Mid**	53.3%	5.84 bpt	63.5%	56.3
4	**DKV-High**	59.1%	12.14 bpt	24.1%	48.6
5	**StreamingLLM**	36.5%	6.00 bpt	62.5%	44.3
6	**Dense FP16** (Reference)	60.0%	16.00 bpt	0.0%	42.0
7	**Low-Rank KV**	28.6%	6.09 bpt	62.0%	38.6
8	**KV-Merging**	19.9%	6.03 bpt	62.3%	32.6

Table 6: Qwen2.5 7B Part 2 ($\mathcal{S}_{\text{sys}}$) Hardware System Serving Leaderboard.

Rank	Method	$\mathcal{S}_{\text{sys}}$	$\mathcal{S}_{\text{res}}$	TTFT	Decode	Peak VRAM	ϕ
1	**KV-Quant**	**72.8**	61.1	5.46s	16.4 tok/s	3854.3 MB	1.00×
2	**StreamingLLM**	61.0	44.3	5.51s	24.9 tok/s	3854.4 MB	1.00×
3	**Dense FP16**	59.4	42.0	5.56s	16.3 tok/s	3854.2 MB	1.00×
4	**Low-Rank KV**	54.9	38.6	6.82s	18.1 tok/s	3854.4 MB	0.93×
5	**DKV-Mid**	54.2	56.3	39.55s	15.3 tok/s	3854.2 MB	0.49×
6	**SnapKV**	53.7	59.5	58.69s	22.2 tok/s	5132.6 MB	0.40×
7	**KV-Merging**	52.8	32.6	5.51s	25.8 tok/s	3854.4 MB	1.00×
8	**DKV-High**	48.5	48.6	41.88s	15.3 tok/s	3854.5 MB	0.48×

B Full Task-Wise Breakdown Tables

B.1 Gemma 4 E2B Task Breakdown (5 Tasks × 5 Context Lengths)

Table 7: Gemma 4 E2B Task-Wise Accuracy Breakdown across Context Lengths (2K to 128K tokens).

Task / Method	2K	8K	32K	64K	128K	Mean Acc
Task 1: Single Needle in a Haystack (Single NIAH)						
KIVI	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
DKV-High	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
Dense FP16	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
SnapKV	91.7%	100.0%	100.0%	100.0%	100.0%	98.3%
DKV-Mid	100.0%	100.0%	100.0%	100.0%	83.3%	96.7%
Low-Rank KV	100.0%	53.3%	50.0%	50.0%	50.0%	60.7%
StreamingLLM	54.9%	50.0%	50.0%	50.0%	50.0%	51.0%
KV-Merging	48.7%	52.1%	42.7%	52.4%	35.7%	46.3%
Task 2: Multi-Needle Extraction (Multi NIAH)						
DKV-High	33.3%	33.3%	33.3%	33.3%	33.3%	**33.3%**
SnapKV	33.3%	33.3%	33.3%	33.3%	27.8%	32.2%
KIVI	33.3%	33.3%	33.3%	33.3%	27.8%	32.2%
Dense FP16	33.3%	33.3%	33.3%	33.3%	27.8%	32.2%
DKV-Mid	33.3%	33.3%	22.2%	11.1%	0.0%	20.0%
Low-Rank KV	33.3%	33.3%	16.7%	5.6%	5.6%	18.9%
StreamingLLM	16.7%	16.7%	16.7%	16.7%	16.7%	16.7%
KV-Merging	16.7%	16.7%	16.7%	0.0%	0.0%	10.0%
Task 3: Key-Value Retrieval (RULER KV)						
SnapKV	33.3%	33.3%	33.3%	0.0%	0.0%	**20.0%**
KIVI	33.3%	33.3%	33.3%	0.0%	0.0%	**20.0%**
DKV-High	33.3%	33.3%	33.3%	0.0%	0.0%	**20.0%**
Dense FP16	33.3%	33.3%	33.3%	0.0%	0.0%	**20.0%**
DKV-Mid	33.3%	0.0%	33.3%	0.0%	0.0%	13.3%
Low-Rank KV	33.3%	16.7%	0.0%	0.0%	0.0%	10.0%
KV-Merging	16.7%	0.0%	0.0%	0.0%	0.0%	3.3%
StreamingLLM	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Task 4: Variable Tracking (RULER Variable Tracking)						
All methods	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Task 5: Multi-Hop Reasoning (Multi-Hop QA)						
DKV-Mid	100.0%	100.0%	100.0%	100.0%	66.7%	**93.3%**
SnapKV	100.0%	100.0%	100.0%	100.0%	16.7%	83.3%
KIVI	100.0%	100.0%	100.0%	100.0%	0.0%	80.0%
DKV-High	100.0%	100.0%	100.0%	100.0%	0.0%	80.0%
Dense FP16	100.0%	100.0%	100.0%	100.0%	0.0%	80.0%
Low-Rank KV	100.0%	100.0%	66.7%	66.7%	0.0%	66.7%
StreamingLLM	83.3%	50.0%	50.0%	50.0%	0.0%	46.7%
KV-Merging	33.3%	0.0%	0.0%	0.0%	0.0%	6.7%

B.2 Gemma 4 E4B Task Breakdown (5 Tasks × 5 Context Lengths)

B.3 Qwen2.5 7B Task Breakdown (5 Tasks × 5 Context Lengths)

Table 8: Gemma 4 E4B Task-Wise Accuracy Breakdown across Context Lengths (2K to 32K tokens).

Task / Method	2K	4K	8K	16K	32K	Mean Acc
Task 1: Single Needle in a Haystack (Single NIAH)						
DKV-High	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
Dense FP16	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
DKV-Mid	100.0%	100.0%	83.3%	83.3%	83.3%	90.0%
KIVI	100.0%	100.0%	100.0%	91.7%	58.3%	90.0%
SnapKV	100.0%	100.0%	100.0%	66.7%	50.0%	83.3%
Low-Rank KV	100.0%	91.7%	60.9%	63.1%	58.8%	74.9%
StreamingLLM	50.0%	52.2%	50.0%	51.1%	50.0%	50.7%
KV-Merging	53.6%	48.5%	38.2%	40.3%	46.6%	45.4%
Task 2: Multi-Needle Extraction (Multi NIAH)						
DKV-Mid	33.3%	33.3%	33.3%	33.3%	33.3%	**33.3%**
SnapKV	33.3%	33.3%	33.3%	33.3%	33.3%	**33.3%**
KIVI	33.3%	33.3%	33.3%	33.3%	33.3%	**33.3%**
DKV-High	33.3%	33.3%	33.3%	33.3%	33.3%	**33.3%**
Dense FP16	33.3%	33.3%	33.3%	33.3%	33.3%	**33.3%**
Low-Rank KV	33.3%	33.3%	22.2%	16.7%	22.2%	25.6%
KV-Merging	16.7%	16.7%	16.7%	22.2%	33.3%	21.1%
StreamingLLM	16.7%	16.7%	16.7%	16.7%	16.7%	16.7%
Task 3: Key-Value Retrieval (RULER KV)						
KIVI	100.0%	100.0%	66.7%	66.7%	100.0%	**86.7%**
Dense FP16	100.0%	100.0%	66.7%	66.7%	100.0%	**86.7%**
SnapKV	66.7%	83.3%	100.0%	83.3%	83.3%	83.3%
DKV-High	100.0%	100.0%	33.3%	33.3%	100.0%	73.3%
Low-Rank KV	100.0%	100.0%	33.3%	33.3%	66.7%	66.7%
DKV-Mid	100.0%	100.0%	33.3%	0.0%	33.3%	53.3%
KV-Merging	33.3%	33.3%	16.7%	16.7%	33.3%	26.7%
StreamingLLM	0.0%	33.3%	0.0%	0.0%	0.0%	6.7%
Task 4: Variable Tracking (RULER Variable Tracking)						
All methods	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Task 5: Multi-Hop Reasoning (Multi-Hop QA)						
DKV-Mid	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
SnapKV	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
KIVI	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
DKV-High	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
Dense FP16	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
Low-Rank KV	100.0%	83.3%	66.7%	50.0%	83.3%	76.7%
StreamingLLM	50.0%	66.7%	50.0%	50.0%	50.0%	53.3%
KV-Merging	33.3%	50.0%	16.7%	50.0%	16.7%	33.3%

Table 9: Qwen2.5 7B Task-Wise Accuracy Breakdown across Context Lengths (2K to 32K tokens).

Task / Method	2K	4K	8K	16K	32K	Mean Acc
Task 1: Single Needle in a Haystack (Single NIAH)						
SnapKV	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
KV-Quant	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
DKV-High	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
Dense FP16	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
DKV-Mid	100.0%	83.3%	66.7%	47.4%	56.9%	70.9%
StreamingLLM	54.0%	54.3%	53.4%	52.1%	33.1%	49.4%
KV-Merging	22.5%	21.5%	16.8%	14.0%	17.0%	18.4%
Low-Rank KV	31.7%	22.9%	12.8%	5.8%	7.4%	16.1%
Task 2: Multi-Needle Extraction (Multi NIAH)						
Dense FP16	33.3%	33.3%	33.3%	33.3%	33.3%	**33.3%**
SnapKV	27.8%	27.8%	33.3%	33.3%	33.3%	31.1%
KV-Quant	27.8%	33.3%	33.3%	22.2%	33.3%	30.0%
DKV-Mid	22.2%	33.3%	33.3%	22.2%	33.3%	28.9%
DKV-High	22.2%	33.3%	33.3%	33.3%	22.2%	28.9%
StreamingLLM	16.7%	16.7%	16.7%	16.7%	16.7%	16.7%
KV-Merging	5.6%	11.1%	11.1%	5.6%	5.6%	7.8%
Low-Rank KV	11.1%	16.7%	0.0%	5.6%	0.0%	6.7%
Task 3: Key-Value Retrieval (RULER KV)						
KV-Quant	100.0%	100.0%	100.0%	100.0%	50.0%	**90.0%**
DKV-Mid	100.0%	100.0%	100.0%	100.0%	33.3%	86.7%
DKV-High	100.0%	100.0%	100.0%	100.0%	33.3%	86.7%
Dense FP16	100.0%	100.0%	100.0%	100.0%	33.3%	86.7%
SnapKV	100.0%	100.0%	83.3%	83.3%	33.3%	80.0%
Low-Rank KV	16.7%	83.3%	83.3%	50.0%	33.3%	53.3%
StreamingLLM	50.0%	50.0%	16.7%	33.3%	33.3%	36.7%
KV-Merging	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Task 4: Variable Tracking (RULER Variable Tracking)						
KV-Quant	0.0%	0.0%	16.7%	0.0%	0.0%	**3.3%**
Other 7 methods	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Task 5: Multi-Hop Reasoning (Multi-Hop QA)						
StreamingLLM	100.0%	100.0%	100.0%	100.0%	0.0%	**80.0%**
SnapKV	100.0%	100.0%	100.0%	100.0%	0.0%	**80.0%**
KV-Quant	100.0%	100.0%	100.0%	100.0%	0.0%	**80.0%**
DKV-Mid	100.0%	100.0%	100.0%	100.0%	0.0%	**80.0%**
DKV-High	100.0%	100.0%	100.0%	100.0%	0.0%	**80.0%**
Dense FP16	100.0%	100.0%	100.0%	100.0%	0.0%	**80.0%**
KV-Merging	100.0%	100.0%	83.3%	83.3%	0.0%	73.3%
Low-Rank KV	100.0%	100.0%	83.3%	50.0%	0.0%	66.7%

C Measurement Protocol & Execution Environment

Hardware Configuration. All evaluations were conducted on dedicated GPU compute nodes equipped with NVIDIA RTX GPUs with 16GB VRAM, CUDA 12.4, and PyTorch 2.4. Model weights were loaded under 4-bit NormalFloat (NF4) quantization via BitsAndBytes to isolate model weight footprint from KV cache dynamics and ensure full evaluation coverage up to 128K context tokens without out-of-memory crashes.

Inference Pipeline. All candidate methods were driven directly from vendored upstream implementations:

- **KIVI / KV-Quant**: Asymmetric per-channel Key quantization and per-token Value quantization.
- **SnapKV**: Attention pooling kernel $k = 7$ with observation window $L_{\text{obs}} = 32$.
- **Low-Rank KV**: Truncated SVD per attention head.
- **StreamingLLM**: 4 initial attention sink tokens with rolling local recency window.
- **KV-Merging**: Temporal adjacent token mean-pooling.

Prefill was executed using chunked forward passes ($C_{\text{chunk}} = 4{,}096$) to avoid PyTorch intermediate tensor allocation spikes. The context memory transform was executed directly on the resident PyTorch KV cache before generating continuation tokens. All decoding used greedy search ($T = 0$) to eliminate sampling variance.

D Proofs and Analysis of Scoring Formulations

Proposition 1 (Monotonicity and Pareto Consistency)**.** Let $\mathcal{S}_{\text{res}}(Q, R) = \alpha Q + (1 - \alpha)R$ with $\alpha \in (0, 1)$. For all $(Q, R) \in [0, 100]^2$, $\mathcal{S}_{\text{res}}$ strictly satisfies Quality Monotonicity, Resource Monotonicity, Boundedness, and Pareto Consistency.

Proof. **1. Monotonicity**: $\frac{\partial \mathcal{S}}{\partial Q} = \alpha > 0$ and $\frac{\partial \mathcal{S}}{\partial R} = 1 - \alpha > 0$.

2. Boundedness: Since $Q, R \in [0, 100]$, the minimum occurs at $(0, 0)$ where $\mathcal{S} = 0$, and the maximum at $(100, 100)$ where $\mathcal{S} = 100\alpha + 100(1 - \alpha) = 100$.

3. Pareto Dominance: If $(Q_A \geq Q_B \wedge R_A \geq R_B)$ with at least one strict inequality, then:

$$\mathcal{S}(Q_A, R_A) - \mathcal{S}(Q_B, R_B) = \alpha(Q_A - Q_B) + (1 - \alpha)(R_A - R_B) > 0. \tag{1}$$

□

E Supplementary Visualizations

Figure 1 illustrates the full distribution of Part 2 hardware system scores across all three foundation models, complementing Tables 2, 4, and 6. Figure 2 provides detailed sensitivity curves for the quality preference parameter α and compares candidate non-linear utility formulations against the canonical CRBench linear metric.

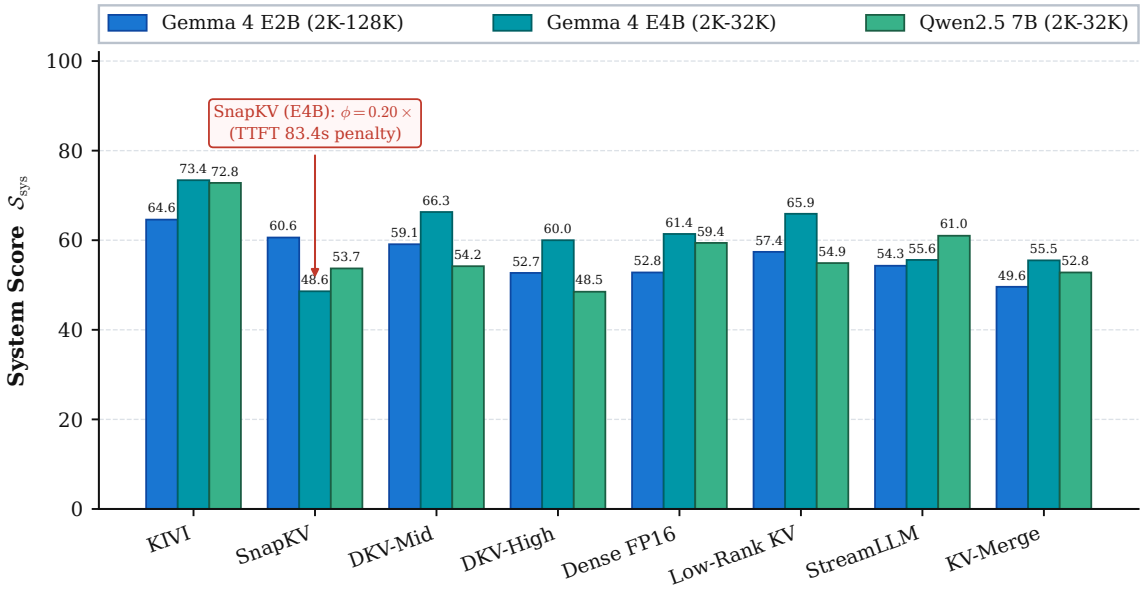


Figure 1: **Cross-Model Part 2 Hardware System Serving Utility Distribution.** Breakdown of $\mathcal{S}_{\text{sys}}$ across Gemma 4 E2B, Gemma 4 E4B, and Qwen2.5 7B.

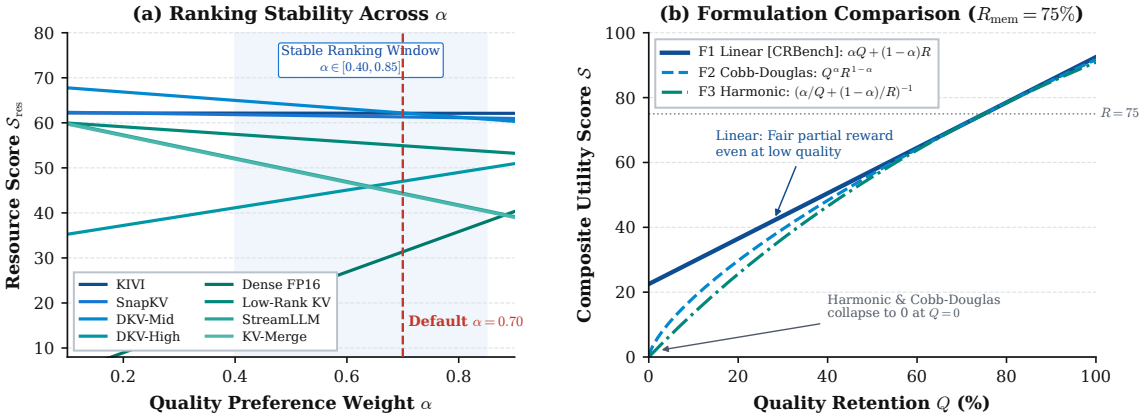


Figure 2: **Sensitivity Analysis and Utility Formulation Comparison.** (a) Ranking trajectories across quality preference weight $\alpha \in [0.1, 0.9]$. (b) Functional behavior of Linear, Cobb–Douglas, and Harmonic utility formulations at fixed memory reduction $R_{\text{mem}} = 75\%$.